

Evaluating the efficacy of non-invasive genetic sampling of the Northern Pacific rattlesnake with implications for other venomous squamates

Brett Ford¹ · Purnima Govindarajulu² · Karl Larsen³ · Michael Russello¹

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Abstract A common challenge for conservation genetic studies is finding minimally invasive sampling methods that maximize the quantity and quality of data produced. Conventional approaches rely on tissue or blood sampling, which typically require lengthy handling times, and can be hazardous for high-risk species, such as venomous snakes. Finding alternative, less invasive techniques is imperative in such circumstances. We compared DNA quantity and genotyping success of blood samples to those from buccal swabs, cloacal swabs, and scale clippings in the Northern Pacific rattlesnake (*Crotalus oreganus oreganus*). Buccal swabs and scales yielded significantly less DNA compared to blood, whereas cloacal samples were not significantly different. Cloacal swabs produced the highest PCR success and lowest genotyping errors commensurate to the blood samples, but differences were not significant between sample types. Our findings suggest cloacal swabbing as an efficient, less invasive alternative for providing high

quality genotypic data for squamate reptiles (venomous or nonvenomous).

Keywords Snake · Microsatellite · Molecular ecology

A thorough understanding of genetics has become fundamental for wildlife management (Frankham 1995). Non-invasive sampling methods (Beja-Pereira et al. 2009) are especially valuable for species of conservation concern (Smith and Wang 2014). With the on-going decline in herpetofauna and continued practice of invasive sampling, alternative sampling methods for amphibians and reptiles warrant investigation. Invasive methods are particularly hazardous for high-risk species, such as venomous snakes, as most snakebites are a consequence of intentional exposure (Morandi and Williams 1997). Buccal (Beebee 2008) and cloacal swabbing (Mucci et al. 2014), and collection of feces (Jones et al. 2008) and shed skins (Fetzner 1999) have all been considered as viable replacements for invasive methods.

The Northern Pacific rattlesnake (*Crotalus oreganus oreganus*) is a venomous species listed as “threatened” under the Species at Risk Act in Canada. Lack of genetic studies has limited knowledge of population connectivity and the ability to identify “designatable units” for management (COSEWIC 2015). To support such initiatives, we compared DNA quantity and genotyping success from scale clippings, buccal swabs, and cloacal swabs to those from more conventional blood samples to determine an effective alternative for obtaining genetic data.

Twelve rattlesnakes were captured in the Okanagan Valley, British Columbia (permit #PE15-171661; Thompson Rivers University Animal Ethics Protocol #100344). Three ventral scutes were clipped from each animal, and placed

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✉ Michael Russello
michael.russello@ubc.ca

¹ Department of Biology, University of British Columbia, Okanagan Campus, 3247 University Way, Kelowna, BC V1V 1V7, Canada

² Ministry of Environment, PO Box 9338, Station Provincial Government, Victoria, BC V8W 9M1, Canada

³ Department of Natural Resource Sciences, Thompson Rivers University, 900 McGill Road, Kamloops, BC V2C 0C8, Canada

into a 1.5 ml microcentrifuge tube. A long-stem sterile swab (Puritan Medical Products, 25-8061WC) was rotated around the outside of the cloacal vent. A 1.0 cc syringe with 25-gauge needle (Terumo Medical Products, cat#SS-01T2516) was inserted into the caudal vein, from which 0.1 ml of blood was drawn. Each snake was placed in a short tube, allowing the head to slightly protrude from the end. While the snake was held at the base of the tube, a long-stem sterile swab was inserted into the back of the mouth. Blood, cloacal and buccal samples were immediately placed into 1 ml Longmire Buffer (Longmire et al. 1997) and all samples were stored at 4 °C.

DNA was extracted from 20 µl of blood-buffer samples using the Nucleospin QuickBlood Kit (Macherey-Nagel, Duren, Germany) and manufacturer's protocols. All other sample types were extracted using the Nucleospin Tissue Kit (Macherey-Nagel, Duren, Germany). Each buccal and cloacal swab was cut in half, placed into a 2 ml microcentrifuge tube with Buffer T1 and Proteinase K, and incubated at 56 °C for 1 h. All three scales were incubated overnight in Buffer T1 and Proteinase K until completely dissolved. Following incubation, subsequent steps followed manufacturer's protocols. DNA was quantified using picogreen and a real-time PCR fluorescence protocol (Blotta et al. 2005) on an Applied Biosystems ViiA7™ (Life Technologies, Carlsbad, CA).

Samples were genotyped at seven microsatellite loci using pre-screened primers [5A, 7-87, 7-150 (Villarreal et al. 1996); MFRD5, MFR12, MFR15, MFR23 (Oyler-McCance et al. 2005)]. Loci were amplified in triplicate

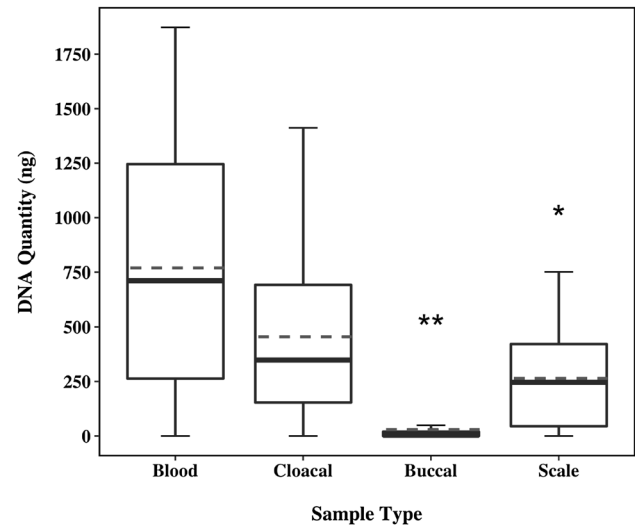


Fig. 1 DNA quantities for *Crotalus oreganus oreganus* sample types: blood, cloacal swabs, buccal swabs, and scales. Dashed lines represent mean values. ** $p < 0.005$; * $p < 0.05$ compared to blood sampling. The bar plot was constructed in R (R Core Team 2016)

for all sample types across all individuals (144 reactions per locus), and fragment analysis was conducted using an Applied Biosystems 3130XL DNA automated sequencer. Alleles were scored in GeneMapper 4.0 (Applied Biosystems, Inc., Foster City, USA). For each sample, we measured: (1) PCR amplification success; (2) allelic dropout (ADO) rate; and (3) false allele (FA) rate using Gimlet 1.3.3 (Valière 2002). “True” genotypes were constructed from blood samples and used as standards for calculating

Table 1 PCR amplification success, allelic dropout (ADO) rate, and false allele (FA) rate across three replicates each for all four sample types at seven microsatellite loci

	MFRD5	MFR12	MFR15	MFR23	7-87	5A	7-150	Mean
Blood samples								
PCR success (%)	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0 ± 0.0
ADO rate (%)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0 ± 0.0
FA rate (%)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0 ± 0.0
Buccal swabs								
PCR success (%)	92.0	92.0	97.0	92.0	100.0	92.0	92.0	93.9 ± 1.2
ADO rate (%)	0.0	5.6	44.4	0.0	0.0	38.1	46.7	19.3 ± 8.5
FA rate (%)	0.0	16.7	0.0	0.0	0.0	0.0	0.0	2.4 ± 2.4
Cloacal swabs								
PCR success (%)	100.0	100.0	100.0	97.0	100.0	97.0	97.0	98.8 ± 0.5
ADO rate (%)	0.0	0.0	0.0	0.0	0.0	4.8	0.0	0.6 ± 0.6
FA rate (%)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0 ± 0.0
Scales								
PCR success (%)	100.0	86.0	100.0	100.0	100.0	100.0	100.0	98.0 ± 1.8
ADO rate (%)	22.2	0.0	0.0	0.0	0.0	9.5	0.0	4.5 ± 3.2
FA rate (%)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0 ± 0.0

genotyping errors. One-way ANOVA was used to compare differences between DNA concentrations between sample types, while two-way ANOVA was used to test differences in genotyping success between samples types and loci. We conducted a post hoc Tukey's test in R (R Core Team 2016) to identify significant differences for all comparisons.

DNA was successfully extracted from all sample types, with mean DNA quantities ranging between 29 and 769 ng (Fig. 1). Buccal swabs and scales yielded significantly lower amounts of DNA compared to blood (Fig. 1, Table S1). Cloacal swabs yielded greater DNA compared to buccal swabs and scales, and less DNA than blood samples, but differences were not significant.

All sampling approaches showed high amplification success rates (>93.9%; Table 1), with buccal swabs producing significantly lower PCR success ($p < 0.05$) compared to blood and cloacal samples (Table 1, Figure S1). Buccal swabs produced the highest genotyping errors compared to other samples, but differences were not significant ($p > 0.05$; Table 1, Table S2).

Blood sampling, while providing robust DNA, is not straightforward; it requires training and experience to be conducted properly. Even when mastered, animals are restrained longer than less-invasive approaches. None of the methods tested here is truly non-invasive, nonetheless, some are less costly and more convenient. Buccal swabbing requires lengthy handling time and extreme caution when working with venomous species, and scale-clipping introduces risk of infection. Cloacal swabbing, however, requires inexpensive equipment, short handling time, and can be conducted by one person, even on venomous species. Based on these considerations and our results, we recommend cloacal swabbing as an effective alternative to blood sampling in Northern Pacific rattlesnakes, with likely utility for studies of other squamates.

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